FOOTYEDGE AI — GEMINI CLI REFACTOR DIRECTIVE (PRODUCTION BLUEPRINT)

You are working on a production betting intelligence system called FootyEdge AI.

Your job is NOT to add more AI usage.

Your job is to RE-ARCHITECT the system into a deterministic + ML-first + minimal-LLM design that aligns with a quant-grade betting intelligence platform.

⚠️ CORE RULE (NON-NEGOTIABLE)

Do NOT use LLMs or prompts for:

- probability calculations
- value bet detection
- odds analysis
- bankroll sizing
- sharp money detection
- feature engineering
- backtesting logic
- match prediction logic
- staking decisions

These must be implemented using:

- deterministic Python logic
- statistical models (Poisson, Elo, probability distributions)
- or trained ML models (LightGBM / XGBoost / sklearn pipelines)

🤖 LLM USAGE IS STRICTLY LIMITED TO:

LLMs are ONLY allowed for:

- human-readable explanations of predictions

- strategy recommendations
- user-facing betting summaries
- chatbot / copilot interaction
- signal narration (formatting only, not decision-making)

LLMs MUST NEVER influence numeric outputs or betting decisions.

🧱 SYSTEM ARCHITECTURE (ENFORCED)

1. Data Layer

Sources:

- API-Football
- Odds APIs (RapidAPI / bookmakers)
- Supabase database
- Local cache layer

Rules:

- caching allowed
- retry + fallback required
- raw provider responses must never be exposed downstream

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2. Feature Engineering Engine (DETERMINISTIC ONLY)

Convert raw football data into ML-ready features.

Must include:

- form metrics (5/10/20 matches)
- home/away performance
- head-to-head stats
- league strength indicators
- player availability/injury impact

RULES:

- fully deterministic transformations only
- NO LLM-generated features
- NO embeddings for decision logic

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3. Prediction Layer (ML + STATISTICAL ONLY)

Allowed models:

- Poisson goal model (mandatory baseline)
- Elo rating system
- LightGBM / XGBoost (optional ML layer)

Outputs:

- home/draw/away probabilities
- expected goals (xG)
- score distributions
- goal market probabilities

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4. Hybrid Probability Engine

Final probability must be computed as:

final_prob = 0.6 × ML_probability + 0.4 × Poisson_probability

Increase Poisson grid range:

- from 10 → 12 goals for higher accuracy

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5. Market Intelligence Layer (DETERMINISTIC)

Core logic:

EV Calculation

EV = (Probability × Odds) − 1

Value Thresholds:

- EV > 0.03 → Value Bet
- EV > 0.07 → Strong Value
- EV > 0.12 → Premium Signal

Sharp Money Detection:

- detect odds movement > 8%
- compare opening vs current odds
- flag abnormal bookmaker divergence

NO LLM USAGE.

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6. Risk Engine (DETERMINISTIC)

Kelly Criterion:

$f^* = (bp - q) / b$

Where:

- b = odds − 1
- p = probability
- $q = 1 - p$

Rules:

- max 25% Kelly cap
- recommended 0.25–0.5 fractional Kelly
- bankroll exposure limits enforced

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7. Backtesting Engine (MANDATORY GATE)

System cannot go live without:

- minimum 5000 historical matches
- ROI calculation
- yield tracking
- drawdown analysis
- profit factor evaluation

NO BACKTEST = NO PRODUCTION RELEASE.

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8. Signal Layer (OUTPUT ONLY)

Responsible ONLY for formatting:

- JSON output
- Telegram formatting
- UI display formatting

STRICT RULE:

- cannot modify probabilities
- cannot modify EV
- cannot alter staking logic

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9. AI LAYER (LLM ISOLATION ZONE ONLY)

All LLM logic must live in:

/ai_explainer/

Responsibilities:

- explain predictions
- summarize bets
- user strategy assistant
- narrative generation

LLM cannot:

- compute odds
- decide bets
- influence EV
- adjust probabilities

🧠 MANDATORY ENFORCEMENT RULES

1. Feature Engineering Rule

All feature engineering must be deterministic statistical transformations.

NO LLM OR AI-GENERATED FEATURES.

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2. ML Training Rule

- all ML models must be trained offline
- only inference allowed in production
- no runtime model generation

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3. Backtesting Rule

No model or system is valid without:

- 5000+ match dataset validation
- ROI + drawdown metrics

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4. Signal Integrity Rule

Signal layer must NEVER:

- modify probability
- modify EV
- modify staking logic

It is a formatting layer ONLY.

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5. LLM Boundary Rule

If any LLM is detected in:

- prediction
- EV calculation

- risk logic

It is INVALID and must be refactored into deterministic logic immediately.

CRITICAL ENFORCEMENT ADDENDUM:

1. Phase 0 must be split into:

   - Repo Map

   - Data Audit

   - Flow Mapping

2. Feature engineering must produce 100–300 deterministic ML features including:

   - form

   - xG proxies

   - league strength

   - tactical mismatch

   - player impact

3. Simulation outputs are forbidden in production pipelines:

   - They must never reach EV, Kelly, backtesting, or signal layers

4. ML pipeline must include:

   dataset_builder → feature_engine → train → evaluate → export

5. All models must be versioned and reproducible

📦 REQUIRED REFACTOR OUTPUT

Gemini must produce:

1. Full refactored folder structure
2. Service architecture breakdown
3. Predictor module rewrite (Poisson + ML hybrid)
4. EV engine implementation
5. Risk (Kelly) engine implementation
6. Signal generator module
7. AI explainer module separation
8. Backtesting engine design

🎯 FINAL SYSTEM GOAL

Transform FootyEdge AI from:

LLM-assisted betting assistant

into:

Quant-grade deterministic betting intelligence platform with ML forecasting and isolated AI explanation layer

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